

```
In [10]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import numpy as np
```

```
In [12]: # Load the data from text files
matrix1 = pd.read_csv('/Users/thejaarladda/Library/CloudStorage/OneDrive-IndianaUniversity/Semester_3/HighThru
matrix1.head()
# To construct 12 * 12 table we need to drop unnamed column containing NaN values
matrix1_col = matrix1.drop(columns=["Unnamed: 13"])
matrix1_col.head()
```

Out[12]:

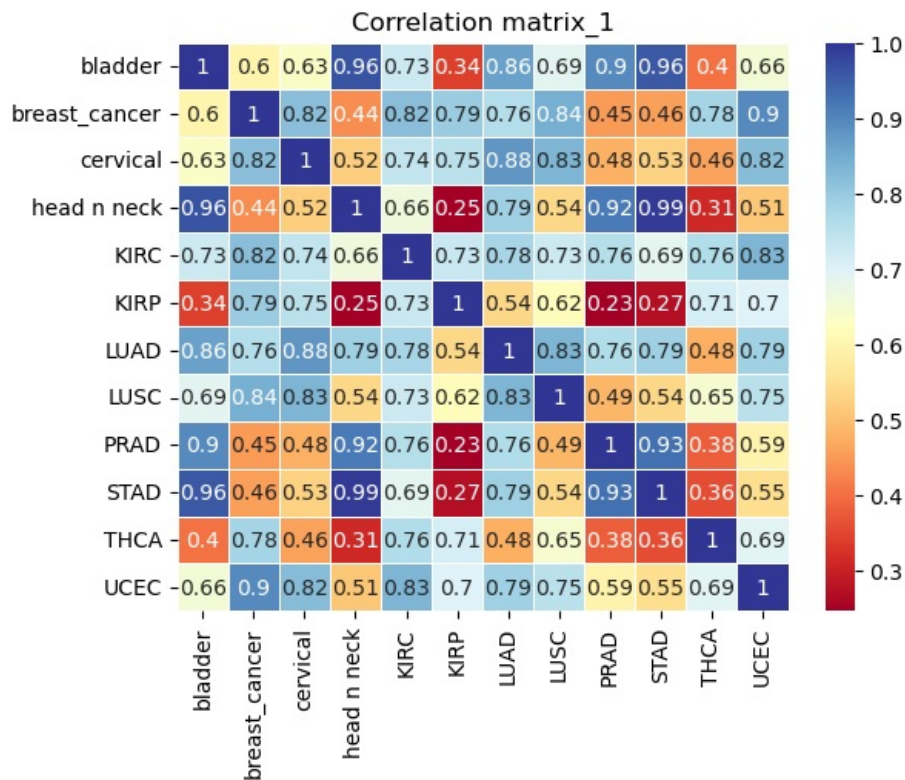
	bladder	breast_cancer	cervical	head n neck	KIRC	KIRP	LUAD	LUSC	
miRNA									
hsa-mir-3128	0.000000	0.000000	0.000000	0.517950	0.000000	0.000000	0.000000	0.000000	(
hsa-mir-3658	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	(
hsa-mir-217	8.155420	13.079703	1.292636	1.553849	22.170186	8.636935	11.046939	13.892229	48
hsa-mir-22	63975.513995	67989.751326	33975.331973	38610.030611	62109.621268	53628.721821	46660.234247	136844.266034	41627
hsa-mir-612	0.107308	0.000000	0.430879	1.035899	0.000000	0.000000	0.000000	0.414693	(

```
In [13]: # Computing a 12 x 12 Pearson correlation matrix for the dataset.
correlation_matrix1 = matrix1_col.corr(method='pearson')
print(correlation_matrix1)
```

	bladder	breast_cancer	cervical	head n neck	KIRC	\	
bladder	1.000000	0.599523	0.634296	0.963368	0.725427		
breast_cancer	0.599523	1.000000	0.817857	0.436347	0.819293		
cervical	0.634296	0.817857	1.000000	0.520650	0.740528		
head n neck	0.963368	0.436347	0.520650	1.000000	0.659066		
KIRC	0.725427	0.819293	0.740528	0.659066	1.000000		
KIRP	0.342192	0.794081	0.750458	0.246662	0.729723		
LUAD	0.862249	0.761040	0.880549	0.790977	0.778040		
LUSC	0.690401	0.841805	0.833315	0.537158	0.725342		
PRAD	0.895310	0.451565	0.478204	0.917496	0.758492		
STAD	0.957167	0.455907	0.528248	0.992209	0.686506		
THCA	0.404031	0.781363	0.460994	0.309508	0.761811		
UCEC	0.655229	0.896039	0.821949	0.510117	0.834949		
	KIRP	LUAD	LUSC	PRAD	STAD	THCA	\
bladder	0.342192	0.862249	0.690401	0.895310	0.957167	0.404031	
breast_cancer	0.794081	0.761040	0.841805	0.451565	0.455907	0.781363	
cervical	0.750458	0.880549	0.833315	0.478204	0.528248	0.460994	
head n neck	0.246662	0.790977	0.537158	0.917496	0.992209	0.309508	
KIRC	0.729723	0.778040	0.725342	0.758492	0.686506	0.761811	
KIRP	1.000000	0.536186	0.618880	0.234074	0.271270	0.713086	
LUAD	0.536186	1.000000	0.832479	0.762357	0.791001	0.475722	
LUSC	0.618880	0.832479	1.000000	0.485356	0.536082	0.645129	
PRAD	0.234074	0.762357	0.485356	1.000000	0.928102	0.380546	
STAD	0.271270	0.791001	0.536082	0.928102	1.000000	0.358905	
THCA	0.713086	0.475722	0.645129	0.380546	0.358905	1.000000	
UCEC	0.697043	0.785303	0.752578	0.594417	0.547637	0.691314	
	UCEC						
bladder	0.655229						
breast_cancer	0.896039						
cervical	0.821949						
head n neck	0.510117						
KIRC	0.834949						
KIRP	0.697043						
LUAD	0.785303						
LUSC	0.752578						
PRAD	0.594417						
STAD	0.547637						
THCA	0.691314						
UCEC	1.000000						

```
In [14]: # Generateing heatmap to visualize the correlation matrices.
sns.heatmap(correlation_matrix1, annot=True, cmap="RdYlBu", linewidths=0.5, robust=True)
```

```
plt.title("Correlation matrix_1")
plt.show()
```



```
In [16]: # Loading Dataset 2 to read
matrix2 = pd.read_csv('/Users/thejaarlaggadda/Library/CloudStorage/OneDrive-IndianaUniversity/Semester_3/HighThru
matrix2.head()

# To construct 12 * 12 table we need to drop unnamed column containing NaN values
matrix2_col = matrix2.drop(columns=["Unnamed: 13"])
matrix2_col.head()
```

Out[16]:

	bladder	breast_cancer	cervical	head n neck	KIRC	KIRP	LUAD	LUSC	
miRNA									
hsa-mir-3128	0.000000	0.000000	0.118438	0.000000	0.000000	0.000000	0.000000	0.298948	(
hsa-mir-3658	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	(
hsa-mir-217	1.771178	25.559522	12.080640	7.977180	1.456229	153.204962	26.312068	13.452661	13
hsa-mir-22	11618.929609	74313.291426	79215.956963	112291.438292	63164.404335	68641.746313	36959.527811	99379.294202	18926
hsa-mir-612	0.000000	0.226190	0.000000	0.000000	0.000000	0.113907	0.000000	0.000000	(

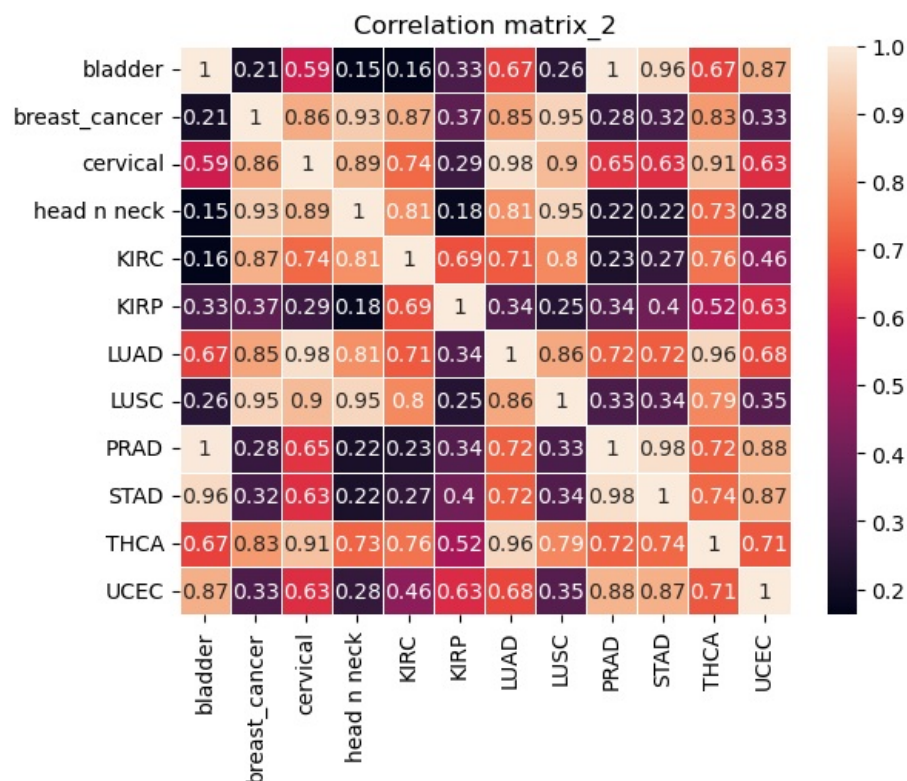
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In [17]: # Computing a 12 x 12 Pearson correlation matrix for the dataset.
correlation_matrix2 = matrix2_col.corr(method="pearson")
print(correlation_matrix2)
```

	bladder	breast_cancer	cervical	head n neck	KIRC	\
bladder	1.000000	0.212732	0.589067	0.149967	0.162164	
breast_cancer	0.212732	1.000000	0.863513	0.931682	0.872940	
cervical	0.589067	0.863513	1.000000	0.885528	0.735477	
head n neck	0.149967	0.931682	0.885528	1.000000	0.808541	
KIRC	0.162164	0.872940	0.735477	0.808541	1.000000	
KIRP	0.326079	0.366644	0.294024	0.181031	0.693256	
LUAD	0.665509	0.847675	0.975662	0.809941	0.709959	
LUSC	0.255758	0.947829	0.896586	0.954687	0.795664	
PRAD	0.996032	0.284830	0.645493	0.220799	0.225755	
STAD	0.963659	0.317559	0.634084	0.223279	0.274682	
THCA	0.672036	0.826725	0.910494	0.727526	0.755754	
UCEC	0.869132	0.333219	0.631045	0.279772	0.461262	

	KIRP	LUAD	LUSC	PRAD	STAD	THCA	\
bladder	0.326079	0.665509	0.255758	0.996032	0.963659	0.672036	
breast_cancer	0.366644	0.847675	0.947829	0.284830	0.317559	0.826725	
cervical	0.294024	0.975662	0.896586	0.645493	0.634084	0.910494	
head n neck	0.181031	0.809941	0.954687	0.220799	0.223279	0.727526	
KIRC	0.693256	0.709959	0.795664	0.225755	0.274682	0.755754	
KIRP	1.000000	0.343165	0.245074	0.343807	0.404890	0.518588	
LUAD	0.343165	1.000000	0.856912	0.719478	0.716984	0.955096	
LUSC	0.245074	0.856912	1.000000	0.326241	0.344390	0.791147	
PRAD	0.343807	0.719478	0.326241	1.000000	0.976378	0.723349	
STAD	0.404890	0.716984	0.344390	0.976378	1.000000	0.738161	
THCA	0.518588	0.955096	0.791147	0.723349	0.738161	1.000000	
UCEC	0.626192	0.676589	0.350942	0.880459	0.874500	0.707618	

	UCEC
bladder	0.869132
breast_cancer	0.333219
cervical	0.631045
head n neck	0.279772
KIRC	0.461262
KIRP	0.626192
LUAD	0.676589
LUSC	0.350942
PRAD	0.880459
STAD	0.874500
THCA	0.707618
UCEC	1.000000

```
In [18]: # Generateing heatmap to visualize the correlation matrices.
sns.heatmap(correlation_matrix2, annot=True, linewidths=0.5, robust=True)
plt.title("Correlation matrix_2")
plt.show()
```



```
In [19]: # Compute Pearson correlation between corresponding columns of Matrix 1 and Matrix 2
correlation_between_matrices = matrix1_col.corrwith(matrix2_col, axis=0, method="pearson")

# Print the correlation in the desired format
print("\nPearson Correlation Between Matrix 1 and Matrix 2:")
```

```
print(correlation_between_matrices)
```

Pearson Correlation Between Matrix 1 and Matrix 2:

bladder	0.877022
breast_cancer	0.914114
cervical	0.891789
head n neck	0.417116
KIRC	0.899953
KIRP	0.605401
LUAD	0.967015
LUSC	0.937687
PRAD	0.915538
STAD	0.962579
THCA	0.654124
UCEC	0.540414

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